Anway Pimpalkar

Cambridge, MA • apimpalkar@seas.harvard.edu ↗ • anway.me ↗

EDUCATION

Harvard University 2025 –
PhD, Bioengineering
Advisors: Katia Bertoldi, PhD (Harvard) and Giovanni Traverso, PhD (MIT)

Johns Hopkins University 2023 – 2025
MSE, Biomedical Engineering
Thesis: Bridging Perception & Performance: Generating, Calibrating, & Evaluating Multimodal Sensory Feedback
Advisor: Jeremy D. Brown, PhD

College of Engineering Pune, India 2019 – 2023
BTech, Electronics and Telecommunication Engineering

AWARDS & HONORS

Harvard SEAS Prize Fellowship ($281,796 • 6-yr tuition, 2-yr stipend) • Harvard University 2025 – 2031

National Defense Science & Engineering Graduate Fellowship ($277,484) • US DOD 2025 – 2028

GEM Associate Fellowship • National GEM Consortium 2025 – 2031

Best Student Paper Finalist [for C2] • International Conference on Rehabilitation Robotics (ICORR) 2025

Start-Up Challenge Finalist ($10,000) • National Institute on Aging of the NIH 2025

Best Paper Award [for C1] • International Conference on Biomedical Engineering Science and Tech. 2023

Mitacs Globalink Research Award ($9,915) • Dalhousie University, Canada 2022

Best Project Award [for O1] • IEEE National Project Competition, India 2021

RESEARCH EXPERIENCE
abbreviated; details at the end of CV

Massachusetts Institute of Technology • *Laboratory for Translational Engineering* ↗ Cambridge, MA
Harvard University • *Bertoldi Solid Mechanics Group* ↗
Doctoral Research Fellow • Advisors: Giovanni Traverso, PhD and Katia Bertoldi, PhD Oct 2025 –

- Bistable Mechanisms for Ingestible and Interactive Devices
- Ingestible Devices for Stress Management

Johns Hopkins University • *Haptics and Medical Robotics Lab* ↗ Baltimore, MD
Graduate Research Assistant • Advisor: Jeremy D. Brown, PhD Jan 2024 – Jun 2025

- Multimodal Visual-Haptic Neurorehabilitation for Post-Stroke Finger Dexterity
- Pneumatactors: Soft Multimodal Haptic Interfaces

Harvard University • *Biohybrid Organs and Neuroprosthetics Lab* ↗ Cambridge, MA
Research Fellow • Advisor: Shriya S. Srinivasan, PhD May – Aug 2024

- Peripheral Nerve Stimulation to Modulate Cancer Tumor Metastasis
- Surgical Approach for Treating Stress Urinary Incontinence

IIT Bombay • *Human Motor Neurophysiology and Neuromodulation Lab* ↗ Mumbai, India
Research Intern • Advisor: Nivethida Thirugnanasambandam, PhD Jan – Aug 2023

- Quantifying Prospective Component of Sense of Agency using Machine Learning

College of Engineering Pune • *Department of Applied Sciences* ↗ Pune, India
Undergraduate Researcher • Advisor: Mahesh H. Shindikar, PhD Aug 2022 – Aug 2023

- Deep Learning-Based MRI Skull Tissue Segmentation

Dalhousie University • *Agricultural Mechanized Systems Group* ↗ Truro, Canada
Mitacs Globalink Research Intern • Advisor: Travis J. Esau, PhD May – Aug 2022

- Automating Harvesters to Reduce Operator Cognitive Load

Queliz Lifetech • *Early Stage Medtech Startup* ↗ Pune, India
Research Intern, Electronics Subsystem Jun – Sep 2021

- Point-of-Care Device for Finger Dexterity Rehabilitation in Acute Burn Patients

JOURNAL PUBLICATIONS

J4 **Pimpalkar A**, Brown JD. *Pneumatactors: Soft 3D-printed interface for co-located transient and continuous tactile stimuli.* In preparation. 2025.

J3 **Pimpalkar A**, West AM, Rai D, Xu J, Brown JD. *Evaluation of multimodal feedback to augment motor control in precision grip.* In preparation. 2025.

J2 Harris C, **Pimpalkar A**, Aggarwal A, Yang J, Chen X, Schmidgall S, Rapuri S, Greenstein JL, Taylor CO, Stevens RD. *Preoperative risk prediction of major cardiovascular events in noncardiac surgery using the 12-lead electrocardiogram: explainable deep learning approach.* Br J Anaesth; 2025 Sep; doi: 10.1016/j.bja.2025.07.085 ↗

Featured in *The Hub at Johns Hopkins*. Article ↗

J1 **Pimpalkar A**, Slepyan A, Thakor NV. *Vibrations at first contact encode object stiffness before grasp completion.* IEEE Sens Lett. 2025 Aug;9(8):1–4. doi: 10.1109/lsens.2025.3584860 ↗

Featured in *New Scientist Magazine*. Article ↗

CONFERENCE PRESENTATIONS † presenter

FULL-LENGTH PAPERS

C4 West AM, Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. *Cross-modal matching is not bidirectional: Implications for human multisensory feedback*. 2026 Jan. In review.

C3 West AM †, **Pimpalkar A**, Xu J, Brown JD. *SenseMatch: Smartphone-based cross-modal matching for accessible perceptual assessments.* 2025 Int IEEE EMBS Conf Neural Eng (NER), 2025 Nov. In print. [Spotlight Poster]

C2 **Pimpalkar A**†, West AM, Xu J, Brown JD. *Optimizing cross-modal matching for multimodal motor rehabilitation.* 2025 IEEE Int Conf Rehabil Robot (ICORR). 2025 May. p. 559–566. doi: 10.1109/icorr66766.2025.11063112 ↗ [Podium Pres] Best Student Paper Award Finalist.

C1 **Pimpalkar A** †, Patole RK, Kamble KD, Shindikar MH. *Performance evaluation of vanilla, residual, and dense 2D U-Net architectures for skull stripping of augmented 3D T1-weighted MRI head scans.* 2023 Int Conf Biomed Eng Sci Tech. 2023 Feb. p. 131–142. doi: 10.1007/978-3-031-54547-4_11 ↗ [Podium Pres] Best Paper Award.

ABSTRACTS AND SHORT PAPERS

A8 West AM, Kasimcan B, **Pimpalkar A**, Xu J, Brown JD. *Cross-domain surpasses within-domain matching*. 2026 Jan. In review.

A7 **Pimpalkar A** †, Rai D, Bartels UJ, Xu J, Brown JD. *Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task.* 2024 Biomed Eng Soc Ann Meet (BMES). 2024 Oct. [Podium Pres, Abs ↗]

A6 Song H † et al. [including **Pimpalkar A**]. *Towards visual function restoration through photoacoustic stimulation.* Invest Ophthalmol Vis Sci. 2024 Jun;65(7):5415. [Podium Pres, Abs ↗]

A5 **Pimpalkar A** †, Ameta P †, Dalia A, Brown JD. *Pneumatactor arrays for high frequency vibrotactile feedback.* 2024 IEEE Haptics Symp. 2024 Apr. [Live Demo, Work-in-Prog Paper]

A4 **Pimpalkar A** †, Ameta P †, Dalia A †, Brown JD, *Pneumatactor arrays for high frequency vibrotactile feedback.* 2024 US Senate AI Cauc Robotics Demo Day. 2024 Apr. [Live Demo]

A3 Harris C †, **Pimpalkar A**, Aggarwal A, Yang P, Chen X, Taylor CO, Greenstein J, Stevens RD. *Surgical risk prediction using an explainable deep learning approach applied to pre-operative 12-lead electrocardiograms.* 2024 JH Med & Eng Res Retr. 2024 Feb. [Poster Pres]

A2 **Pimpalkar A** †, Patole R, Thirugnanasambandam N. *Demonstrating the prospective component of sense of agency using machine learning.* 2023 COEP ENTC BTech Proj Symp. 2023 May. [Poster Pres]

A1 **Pimpalkar A** †, Patole R, Kamble K, Shindikar M. *Evaluating U-Nets for skull stripping of augmented T1-weighted MRI scans.* 2023 No Garland Neurosci Conf. 2023 Feb. [Podium Pres]

NON PEER-REVIEWED ARTICLES

O1 **Pimpalkar A**, Niture D. *Towards contactless elevators with tinyML using person detection and keyword spotting.* 2024 Jul. Best Project Award. doi: arXiv:2405.13051 ↗

TEACHING ENGAGEMENTS

Principles of the Design of Biomedical Instrumentation • Johns Hopkins University 580.771 Fall 2024
Teaching Assistant to Nitish V. Thakor, PhD

- Mentored labs on designing ECG, EMG, EEG circuits and projects including accessible gaming, wearables.

Haptic Interface Design for Human-Robot Interaction • Johns Hopkins University 530.691 Fall 2024
Class Researcher with Jeremy D. Brown, PhD

- Helped implement an online learning platform providing self-paced lectures incorporating live instructor interaction.

Mechatronics • Johns Hopkins University 530.421 Spring 2024
Teaching Assistant to Jeremy D. Brown, PhD

- Solved a 3-year challenge in real-time location polling of autonomous air hockey robots using CV-based ArUco tag detection and ZigBee mesh networking. Led labs on building autonomous robots. GitHub ↗

INVITED LECTURES

T1 Neural & Rehabilitation Engineering Course, Johns Hopkins University. (Nitish V. Thakor, PhD) [*Haptics: Neural Basis, Applications, Future Avenues*], Sep 2024.

ACADEMIC SERVICE

IEEE Haptics Symposium • Student Volunteer 2024

Canadian Society of Bioengineering Annual Meeting • Student Volunteer 2022

LEADERSHIP AND TEAMWORK

Johns Hopkins India Institute • Operations Team 2024 – 2025

- Managed on-day operations and logistics for the first annual Hopkins-India Conference.

Johns Hopkins Biomedical Engineering • Master’s Program Representative 2024 – 2025

COEP Impressions Cultural Festival • Head of Events (2021), Volunteer 2019 – 2021

- Led the planning and execution of 28 annual events with 10k attendees, managing a ₹2m budget and coordinating with an overall team of 126 members to execute concerts, competitions, and international celebrity engagements.
- Built and led a team of 15, working with 2 co-heads to build a platform from the ground up, fostering opportunities for budding artists across India and uniting them under a shared vision.

Rowing Team • COEP, Pune City • Competitive 1X, 2X, 4X Sculler 2020 – 2022

- Won multiple race events and an institutional award. Represented city at regional races.

COEP Mental Health & Wellness Center • Student Volunteer 2020 – 2022

COEP Data Science & AI Club • Research Lead (2020-21), Member 2020 – 2022

RESEARCH EXPERIENCE detailed

Massachusetts Institute of Technology • *Laboratory for Translational Engineering* ↗ Cambridge, MA
Harvard University • *Bertoldi Group* ↗
Doctoral Research Fellow • Advisors: Giovanni Traverso, PhD and Katia Bertoldi, PhD Sep 2025 –

Bistable Mechanisms for Ingestible and Interactive Devices

Ingestible Devices for Stress Management

Johns Hopkins University • *Haptics and Medical Robotics Lab* ↗ Baltimore, MD
Graduate Research Assistant • Advisor: Jeremy D. Brown, PhD Jan 2024 – Jun 2025

Multimodal Visual-Haptic Neurorehabilitation for Post-Stroke Finger Dexterity

- Developed and evaluated a multimodal feedback framework for motor rehabilitation.
- Integrated cross-modal perceptual calibration to optimize sensory feedback delivery to individual thresholds.
- Implemented on ubiquitous devices to support smartphone-based rehabilitation outside the lab.
- *Publications:* J3, C2 (Best Student Paper Finalist at ICORR 2025), C3, C4, A7, A8

Pneumatactors: Soft Multimodal Haptic Interfaces

- Engineered *"pneumatactors"* – a novel pneumatic tactor haptic interface designed for co-located stimulation of fast and slow adapting mechanoreceptors, enhancing cognitive salience. 3D-printed with flexible TPE to match mechanical impedance of human skin for intuitive interaction.
- *Publications:* J4, A4, A5

Harvard University • *Biohybrid Organs and Neuroprosthetics Lab* ↗ Cambridge, MA
Research Fellow • Advisor: Shriya S. Srinivasan, PhD May – Aug 2024

Peripheral Nerve Stimulation to Modulate Cancer Tumor Metastasis

- Developed non-invasive peripheral nerve stimulation strategies to modulate tumor growth and metastasis in murine models, revealing frequency-dependent effects on tumor innervation. Designed and conducted pilot studies on melanoma-bearing mice.
- Engineered miniaturized, RF-powered vagus nerve stimulators for freely moving rodents to study glioblastoma and nervous system interactions. Devices delivered monophasic pulses up to 1 kHz via custom PCB oscillators, enabling targeted parasympathetic modulation in vivo.

Surgical Approach for Treating Stress Urinary Incontinence

- Innovated and tested a fully endogenous muscle graft-based surgical technique for treating stress urinary incontinence, featuring a 30-minute procedure with potential for fully laparoscopic implementation and superior efficacy compared to urethral devices.
- Conducted pilot surgeries on rat models, synchronizing bladder pressure, flow, and tension using a custom-designed PCB to assess and analyze urodynamic parameters.

IIT Bombay • *Human Motor Neurophysiology and Neuromodulation Lab* ↗ Mumbai, India
Research Intern • Advisor: Nivethida Thirugnanasambandam, PhD Jan – Aug 2023

Quantifying Prospective Component of Sense of Agency using Machine Learning

- Characterized pre-movement EEG signatures underlying the Sense of Agency using a modified Libet clock task with probabilistic auditory feedback. Applied spectral, time-series, and machine learning analyses to identify spatiotemporal patterns predictive of intentional binding.
- Demonstrated distinct neural correlates in pre-SMA and centroparietal regions, supporting a prospective model of agency.
- *Publications:* A2, Bachelor's Dissertation

College of Engineering Pune • *Department of Applied Sciences* ↗ Pune, India
Undergraduate Researcher • Advisor: Mahesh H. Shindikar, PhD Aug 2022 – Aug 2023

Deep Learning-Based MRI Skull Tissue Segmentation

- Developed and benchmarked vanilla, residual, and dense 2D U-Net architectures for skull stripping in neuroimaging, achieving 99.75% accuracy with a dense model optimized for feature reuse. Implemented data augmentation to improve generalizability across multi-scanner datasets. Demonstrated utility of open deep learning pipelines for replacing proprietary tools in MRI workflows.

- *Publications:* C1 (Best Paper Award at ICBEST 2023), A1

Dalhousie University • *Agricultural Mechanized Systems Group* ↗ Truro, Canada

Mitacs Globalink Research Intern • Advisor: Travis J. Esau, PhD May – Aug 2022

Automating Harvesters to Reduce Operator Cognitive Load

- Engineered a closed-loop control system to automate bin switching and dynamic harvester head height adjustment, reducing operator load. Used RGBD sensing for volumetric bin fill estimation (±10% accuracy), and integrated crop height and GPS data to actuate harvester head in real time.

Queliz Lifetech • *Early Stage Medtech Startup* ↗ Pune, India

Research Intern, Electronics Subsystem Jun – Sep 2021

Point-of-Care Device for Finger Dexterity Rehabilitation in Acute Burn Patients

- Designed an affordable, portable rehabilitation device to restore MCP, DIP, and PIP joint mobility in burn patients. Developed a precision actuation system with adaptive feedback to enable customized therapy in low-resource clinical settings.